MIL-C-50513B (AR)
AMENDMENT 5
5 September 1984
SUPERSEDING
AMENDMENT 4
16 November 1982

MILITARY SPECIFICATION

CARTRIDGE, IGNITION, M2A2
PARTS FOR, LOADING, ASSEMBLING AND PACKING

This amendment forms a part of Military Specification MIL-C-50513B (AR) dates 27 February 1981, and is approved for use by the U.S. Army Armament Munitions and Chemical Command and is available for use by all Departments and Agencies of the Department of Defense.

PAGE 1

2.1, Specifications, Military: Add "MIL-P-116 Preservation-Packing, Method of"

PAGE 2

2.1, Standards, Military, add:

"MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective."

PAGE 3

* 3.6, Delete paragraph in its entirety and substitute:

"3.6 Gross weight of assembly.

a. With steel head the gross weight of the assembly shall be 368 grains, min.

b. With brass head the gross weight of the assembly shall be 340 grains, min."

PAGE 4

Add new para. 3.9:

"3.9 Burst pressure. The ignition cartridge assembly shall comply with the following requirements when tested as specified in 4.5.5.

FSC 1315



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3.9.1 At +125° Fahrenheit (F). The burst pressure shall not be greater than 4750 pounds per square inch (psi) nor less than 3700 psi.

3.9.2 At +70°F. The burst pressure shall not be greater than 5050 psi nor less than 3800 psi.

3.9.3 At -40°F. The burst pressure shall not be greater than 5300 psi nor less than 3900 psi."

4.3.1, Cartridge, Ignition, M2A2, Quality column, delete: "111" and substitute "261."

PAGE 5

Table I, Cartridge, Ignition, M2A2 Sample column, delete (two places): "111" and substitute: "261."

PAGE 6

Table I, Leak test, Sample column, delete: "100" and substitute: "250"

Add new tests:

Burst test

| | | | |
|---|---|---|---|
| +125°F | 50 | 3.9.1 | 4.5.5 |
| +70°F | 50 | 3.9.2 | 4.5.5 |
| -40°F | 50 | 3.9.3 | 4.5.5 |

PAGE 12

4.4.2.5, Title: add "If applicable"

PAGE 15

4.4.2.8, NOTES: delete "Samples shall be selected at random from each lot at inspection level S-4 under MIL-STD-105 or at inspection level I for use under multilevel sampling plan CSP-M of MIL-STD-1235" and substitute "Sampling, acceptance criteria and test procedure for this test shall be as specified in MIL-P-116."



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4.5.3, line 6: delete "5 sec." and substitute "3 sec."

line 9: delete "0.25 psi" and substitute "0.15 psi"



4.5.4, delete paragraph in its entirety and substitute:

"4.5.4 Proving ground. This test shall be performed at a government proving ground. The ignition cartridge with inert loaded M329A2 projectile shall be assembled into a M329A2 cartridge to a weight of 22.12 plus or minus .01 pounds. The sample shall be fired at the five increment charge.

Each round in the sample shall contain a bag loading assembly (minimum charge) and a five increment bundle. The bag loading assembly shall be centered over the flash holes on the cartridge container and the five increment bundle shall be placed forward of and against the bag. The bag loading assembly and propellant increment bundle shall be from an acceptance stock lot. The rounds shall be fired from a 4.2 Inch M30 mortar at an elevation of 900 mils.

Conditioning rounds will be assembled from stock components and fired to condition the weapon, to check-out instrumentation and spot the field. The number of conditioning rounds required for the temperature conditions specified are as follows:

| Ambient Temp. | Number of Rounds |
|---|---|
| Above 72°F | 2 |
| 32°F to 72°F | 3 |
| Below 32°F | 4 |

Individual muzzle velocities and standard deviation of the sample shall be recorded and reported for information purposes."

Add new para. 4.5.5:

"4.5.5 Burst pressure. The ignition cartridges shall be temperature conditioned at the specified temperatures plus or minus 5°F for a minimum (min.) of three hours. The ignition cartridges shall be removed from the temperature conditioning environment, assembled into a test fixture that simulates the cartridge container (dwg. 9935185) mounted on the 4.2 Inch Mortar Cartridge, fired within one minute, maximum (max.) and



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the peak pressure shall be measured at one of the center holes in the simulated cartridge container. The ignition cartridge to be modified by incorporating a .3 ± .030 inch, diameter hole aligned with the pressure tapped hole in the simulated cartridge container. The hole in the ignition cartridge to be sealed with transparent tape. All the holes in the simulated cartridge container with the exception of the one center hole (tapped for pressure measurement) shall be open to the atmosphere. The first article sample shall be rejected if the burst pressure at each temperature fails to comply with the acceptance criteria of MIL-STD-414, Table B-3, Code M and an AQL of 1.0 percent."

PAGE 20

5.3.1, under 25-CARTRIDGE, IGNITION, M2A2: delete
"LOT (MFR SYMBOL NO.)
LOADED (MO/YR)"
and substitute "LOT NUMBER (See 4.4.1 of MIL-A-48078)"

Add the following paragraphs:

"6.5 Check test for deterioration (see 6.6). If the total elasped time between original acceptance of the ignition cartridge lot and the assembly of that lot into the complete round exceeds two years, or if the ignition cartridges have been subjected to adverse conditions for any period of time since previous tests, the ignition cartridges shall be subjected to, and must satisfactorily comply with the following tests immediately before the ignition cartridges are assembled into the complete round.

6.5.1 Ignition cartridge non-functioning/functioning. The sampling plan for these tests shall be in accordance with MIL-STD-105, Table II A, Code Letter K and an AQL of 0.10 percent for non-functioning and an AQL of 0.40 percent for functioning. The tests shall be performed as specified in 4.5.2.

6.5.2 Moisture content of black powder. Twenty five samples of black powder shall be selected from the ignition cartridges subjected to the non-functioning test specified in 6.5.1. The moisture content shall not exceed 0.50 percent when determined as specified in 4.5.1.

6.5.3 Proving ground. The sampling plan shall be in accordance with MIL-STD-105, Table II A, Code Letter H and an AQL of 1.0 percent. The test shall be performed as specified in 4.5.4.



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6.6 Cost of check test. The contracting officer will arrange for the contractor to be reimbursed for the expense incurred in performance of the check test. The test shall be conducted at government expense without cost to the contractor who loaded the ignition cartridges in the complete rounds and shall not constitute a basis of rejection against either contractor except where deterioration has occurred as a direct result of carelessness in handling, storage, etc. permitted while the primer lots were under the jurisdiction of either contractor."

NOTE: The margins of this amendment are marked with an asterisk or vertical line to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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